Problem D. Reachability Query

Time Limit 3000 ms

Problem Statement

You are given an undirected graph with N vertices and zero edges.
The vertices are numbered $1, 2, \ldots, N$, and initially all vertices are white.
Process a total of Q queries of the following three types:

- Type 1 : Add an undirected edge connecting vertices u and v.
- Type 2 : If vertex v is white, change it to black; if it is black, change it to white.
- Type 3 : Determine whether a black vertex can be reached from vertex v by traversing zero or more edges; report `Yes` if reachable, and `No` otherwise.

Constraints

- All input values are integers.
- $1 \leq N \leq 2 \times 10^5$
- $1 \leq Q \leq 6 \times 10^5$
- Type 1 queries satisfy the following constraints:
  - $1 \leq u < v \leq N$
  - For each query, no edge connecting u and v has been added before that query.
- Type $2, 3$ queries satisfy the following constraints:
  - $1 \leq v \leq N$

Input

The input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \quad Q \\
\text{Query}_1 \\
\text{Query}_2 \\
\vdots \\
\text{Query}_Q
\end{array}
$$

where Query_i represents the i-th query.

Type 1 queries are given in the following format:

```
1 u v
```

Type 2 queries are given in the following format:

```
2 v
```

Type 3 queries are given in the following format:

```
3 v
```

Output

For each type 3 query, output the answer as follows:

- `Yes` if a black vertex can be reached from vertex v by traversing zero or more edges;
- `No` if no black vertex can be reached from vertex v by traversing zero or more edges.

Sample 1

Input	Output
5 12 3 2 2 2 3 2 1 2 5 1 3 4 3 4 3 5 1 4 5 1 1 3 3 1 2 2 3 1	No Yes No Yes Yes No

In this input, the graph initially has five vertices and zero edges.
This input contains 12 queries.

- The 1st query is `3 2`.
  - At this point, no black vertex can be reached from vertex 2 by traversing zero or more edges, so report `No`.
- The 2nd query is `2 2`.
  - Vertex 2 is white, so change it to black.
- The 3rd query is `3 2`.

  - At this point, black vertex 2 can be reached from vertex 2 by traversing zero or more edges. Therefore, report `Yes`.
- The 4th query is `1 2 5`.
  - Add an edge connecting vertices 2, 5.
- The 5th query is `1 3 4`.
  - Add an edge connecting vertices 3, 4.
- The 6th query is `3 4`.
  - At this point, no black vertex can be reached from vertex 4 by traversing zero or more edges, so report `No`.
- The 7th query is `3 5`.
  - At this point, black vertex 2 can be reached from vertex 5 by traversing zero or more edges. Therefore, report `Yes`.
- The 8th query is `1 4 5`.
  - Add an edge connecting vertices 4, 5.
- The 9th query is `1 1 3`.
  - Add an edge connecting vertices 1, 3.
- The 10th query is `3 1`.
  - At this point, black vertex 2 can be reached from vertex 1 by traversing zero or more edges. Therefore, report `Yes`.
- The 11th query is `2 2`.
  - Vertex 2 is black, so change it to white.
- The 12th query is `3 1`.
  - At this point, no black vertex can be reached from vertex 1 by traversing zero or more edges, so report `No`.